

van die invulblads.

University of Pretoria - Department of Mathematics and Applied Mathematics
Universiteit van Pretoria - Departement Wiskunde en Toegepaste Wiskunde

MATHEMATICS WTW158 / WISKUNDE WTW158
SEMESTER TEST 2 / SEMESTERTOETS 2

Date / Datum: May / Mei 2018 Marks / Punte : 50 Time / Tyd : $2\frac{1}{2}$ hours / ure

MARKS FOR SECTIONS B & C PUNTE VIR AFDELINGS B & C	
	35

SURNAME / VAN

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FIRST NAMES / VOORNAME

STUDENT NUMBER

STUDENTENOMMER

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SIGNATURE

HANDTEKENING

CIRCLE YOUR LECTURE GROUP NUMBER / OMKRING JOU LESINGGROEPNOMMER

1	2	3	4	10
Dr Jooste	Prof Harding	Dr Garba	Dr van der Hoff	Mev Dreyer

READ THE FOLLOWING INSTRUCTIONS

1. The paper consists of pages 1 to 11. Check whether your paper is complete.
2. No calculator may be used.
3. No question paper may be removed from the venue.
4. Any queries about the marking must be done within three days after the tests have been handed back.

LEES DIE VOLGENDE INSTRUKSIES

1. Die vraestel bestaan uit bladsye 1 tot 11. Kontroleer of u vraestel volledig is.
2. Geen sakrekenaar mag gebruik word nie.
3. Geen vraestel mag uit die lokaal geneem word nie.
4. Enige navrae oor die nasienwerk moet binne drie dae nadat die toetse teruggegee is, gedoen word.

SECTION A: MULTIPLE CHOICE QUESTIONS - (15 Marks)

1. The questions in Section A must be completed on SIDE 2 of the optical reader form in soft pencil.
2. First circle your answers on this paper and then transfer these to the optical reader form.

AFDELING A : MEERVOUDIGE KEUSEVRAE - (15 Punte)

1. Die vrae in Afdeling A moet op KANT 2 van die merkleesvorm met 'n sagte potlood voltooi word.
2. Omkring eers jou antwoorde op hierdie vraestel en dra dit dan oor na die merkleesvorm.

Question 1 / Vraag 1

If / As $f(x) = \cosh(\ln 2x)$ then / dan is $f'(x) =$

(a)	$-\sinh(\frac{1}{2x})$	(b)	$\sinh(\frac{1}{x})$	(c)	$-\sinh(\ln 2x) \times \frac{1}{x}$
(d)	$\sinh(\ln 2x) \times \frac{1}{x}$	(e)	$\sinh(\ln 2x) \times \frac{1}{2x}$	(f)	None of these / Geen van hierdie

Question 2 / Vraag 2

If / As $f(x) = \sin 3x + 3^x$ then / dan is $f''(x) =$

(a)	$-\sin 3x + 3^x(\ln 3)^2$	(b)	$-9 \sin 3x + 3^x(\ln 3)^2$	(c)	$-9 \sin 3x + 3^x(\ln 9)$
(d)	$-\sin 3x + 3^x(\ln 9)$	(e)	$-9 \sin 3x + 3^x(\ln 3)$	(f)	None of these / Geen van hierdie

Question 3 / Vraag 3

Given / Gegee: $f(x) = \frac{3x^2 - 1}{x^2 - 3x + 2}$

The equation(s) of all horizontal asymptotes is/are
Die vergelyking(s) van alle horisontale asimptote is

(a)	$x = 1$ and / en $x = 2$	(b)	$y = 1$ and / en $y = 2$	(c)	$x = -3$
(d)	$x = 3$	(e)	$y = 3$	(f)	None of these / Geen van hierdie

Question 4 / Vraag 4

Suppose f has a local maximum or minimum at the point c . What theorem guarantees that $f'(c) = 0$ if $f'(c)$ exists?

Veronderstel f het 'n lokale maksimum of minimum in die punt c . Watter stelling waarborg dat $f'(c) = 0$ indien $f'(c)$ bestaan?

(a)	Mean Value Theorem / Middelwaardestelling
(b)	Fermat's theorem / Fermat se stelling
(c)	Extreme Value Theorem / Ekstreemwaardestelling
(d)	Rolle's theorem / Rolle se stelling
(e)	None of these / Geen van hierdie

Question 5 / Vraag 5

$$\lim_{\theta \rightarrow 0} \frac{\sin(5\theta)}{3\theta + \sin \theta} =$$

(a) $\frac{5}{8}$	(b) $\frac{8}{5}$	(c) $\frac{3}{5}$	☒ (d) $\frac{5}{4}$	(e) $\frac{4}{5}$	(f) None of these / Geen van hierdie
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Question 6 / Vraag 6

$$\text{Consider / Beskou } f(x) = \begin{cases} c & \text{if / as } x = 3 \\ \frac{x^2 + 2x - 15}{x - 3} & \text{if / as } x \neq 3 \end{cases}$$

Determine the value of c for which f will be continuous on R .
Bepaal die waarde van c waarvoor f kontinu is op R .

(a) 3	(b) 0	(c) 11	☒ (d) 8	(e) 9	(f) None of these / Geen van hierdie
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Question 7 / Vraag 7

If $f(1) = 4$ and $f'(x) \geq 2$ for $1 \leq x \leq 4$, what is the smallest possible value of $f(4)$?

Indien $f(1) = 4$ en $f'(x) \geq 2$ vir $1 \leq x \leq 4$, wat is die kleinste moontlike waarde van $f(4)$?

(a) 6	(b) 4	(c) 8	☒ (d) 10	(e) $\frac{9}{2}$	(f) None of these / Geen van hierdie
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Question 8 / Vraag 8

$$\lim_{x \rightarrow 0^+} e^{-\frac{1}{x}} =$$

☒ (a) 0	(b) $-\infty$	(c) 1	(d) ∞	(e) $\frac{1}{e}$	(f) None of these / Geen van hierdie
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Question 9 / Vraag 9

The point on the curve $y = \frac{e^x}{x^2 + 1}$ at which the tangent line is horizontal is

Die punt op die kromme $y = \frac{e^x}{x^2 + 1}$, waar die raaklyn horisontaal is, is

(a) $(\frac{1}{2}, 0)$	☒ (b) $(1, \frac{e}{2})$	(c) $(-1, \frac{1}{2e})$
(d) $(0, \frac{1}{2})$	(e) None of these / Geen van hierdie	

Question 10 / Vraag 10

Suppose $f'(x) > 0$ and $f''(x) < 0$ on a given interval I , then f is

Veronderstel $f'(x) > 0$ en $f''(x) < 0$ op 'n gegewe interval I , dan is f

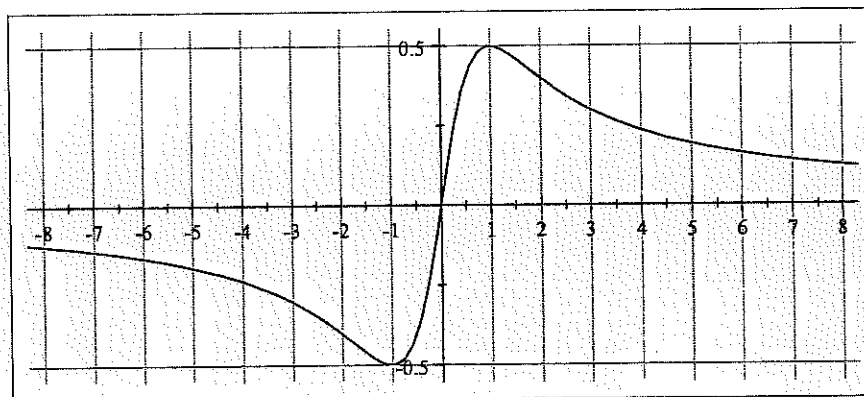
(a)	increasing and concave up on I / stygend en konkaaf na bo op I
☒ (b)	increasing and concave down on I / stygend en konkaaf na onder op I
(c)	decreasing and concave down on I / dalend en konkaaf na onder op I
(d)	decreasing and concave up on I / dalend en konkaaf na bo op I
(e)	None of these / Geen van hierdie

SECTION B (14 Marks) / AFDELING B (14 Punte):

1. The questions in section B must be completed on this paper in pen.
 2. Only the answer in the provided box will be marked.
 3. Calculations can be performed on the facing empty pages and will not be marked.
1. Die vrae in afdeling B moet op die vraestel voltooi word in pen.
 2. Slegs die antwoord in die gegewe blok sal gemerk word.
 3. Berekeninge kan op die oop teenblaai gedoen word en sal nie gemerk word nie.

Question 11 / Vraag 11 (2, 1)

The graph of f' is given. / Die grafiek van f' word gegee.



11.1 Complete the following:

Voltooi die volgende:

f has a local minimum at $x = \dots 0 \dots$

f het 'n lokale by $x = \dots$

11.2 Write down the interval(s) on which f is concave up.

Skryf die interval(le) neer waarop f konkaaf na bo is.

$(-1, 1)$ (or $[-1, 1]$)

Question 12 / Vraag 12 (1)

The critical number of $f(x) = x \ln x - x$ on the interval $(0, \infty)$ is

Die kritieke getal van $f(x) = x \ln x - x$ op die interval $(0, \infty)$ is

$c = 1$ (or $x = 1$)

Question 13 / Vraag 13 (2)

The equation of the tangent line to the graph of $f(x) = \cos 3x$ at the point $x = \frac{\pi}{2}$ is

Die vergelyking van die raaklyn aan die grafiek van $f(x) = \cos 3x$ in die punt $x = \frac{\pi}{2}$, is

$$y = 3\left(x - \frac{\pi}{2}\right) \quad \left(\text{or } y = 3x - \frac{3\pi}{2}\right)$$

Question 14 / Vraag 14 (1)

Give the number c of the Mean Value Theorem for $f(x) = \cos x + x$ on $[0, 2\pi]$.

Gee die getal c van die Middelwaardestelling vir $f(x) = \cos x + x$ op $[0, 2\pi]$.

$$c = \pi$$

Question 15 / Vraag 15 (2, 2, 2)

Give the derivatives of the following. Do not simplify your answer.

Gee die afgeleides van die volgende. Moenie jou antwoord vereenvoudig nie.

15.1 $f(x) = x^2 \ln(x^2 + x)$

$$2x \cdot \ln(x^2 + x) + x^2 \cdot \frac{2x + 1}{x^2 + x}$$

15.2 $f(x) = \arcsin(1 - x^2) = \sin^{-1}(1 - x^2)$

$$\frac{-2x}{\sqrt{1 - (1 - x^2)^2}}$$

15.3 $f(x) = \operatorname{cosec}(\ln x)$

$$-\operatorname{cosec}(\ln x) \cdot \cot(\ln x) \cdot \frac{1}{x}$$

Question 16 / Vraag 16 (1)

If / Indien $a > 0$, $\lim_{x \rightarrow a} \frac{\sqrt{x} - \sqrt{a}}{x - a} =$

$$\frac{1}{2\sqrt{a}}$$

SECTION C (21 Marks) / AFDELING C (21 Punte):

1. The questions in section C must be completed on this paper.
2. All calculations must be shown and will be marked.
3. If you need more space for an answer, use the facing page and indicate it clearly. Also frame the answer on the facing page.
4. Attention should be paid to writing in a mathematically sound way.

1. Die vrae in afdeling C moet op die vraestel voltooi word.
2. Alle berekeninge moet getoon word en sal gemerk word.
3. As jy nog ruimte vir 'n antwoord nodig het, gebruik die teenblad en dui dit duidelik aan. Trek ook 'n raam om die antwoord op die teenblad.
4. Maak seker dat jou skryfwyse wiskundig korrek is.

Question 17 / Vraag 17 (2, 2)

17.1 Given / Gegee: $y = (\arctan x)^x = (\text{bgtan } x)^x = (\tan^{-1} x)^x$. Determine / Bepaal $\frac{dy}{dx}$.

$$y = (\arctan x)^x$$

$$\Rightarrow \ln y = x \ln(\arctan x)$$

$$\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = \ln(\arctan x) + x \cdot \frac{\frac{1}{1+x^2}}{\arctan x}$$

$$\Rightarrow \frac{dy}{dx} = y \left(\ln(\arctan x) + \frac{x}{(1+x^2)\arctan x} \right)$$

$$\Rightarrow \frac{dy}{dx} = (\arctan x)^x \left[\ln(\arctan x) + \frac{x}{(1+x^2)\arctan x} \right]$$

17.2 Determine $\frac{dy}{dx}$ if $e^{x+y} = xy$. / Bepaal $\frac{dy}{dx}$ as $e^{x+y} = xy$.

$$\frac{d}{dx}(e^{x+y}) = \frac{d}{dx}(xy)$$

$$\Rightarrow e^{x+y} \left(1 + \frac{dy}{dx}\right) = y + x \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} (e^{x+y} - x) = y - e^{x+y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y - e^{x+y}}{e^{x+y} - x}$$

Question 18 / Vraag 18 (2)

Prove that / Bewys dat $\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$.

$$\frac{d}{dx} \cot x = \frac{d}{dx} \frac{\cos x}{\sin x}$$

$$= \frac{-\sin x \cdot \sin x - \cos x \cdot \cos x}{\sin^2 x}$$

$$= \frac{-(\sin^2 x + \cos^2 x)}{\sin^2 x}$$

$$= \frac{-1}{\sin^2 x}$$

$$= -\operatorname{cosec}^2 x.$$

Question 19 / Vraag 19 (2)

Use the squeeze theorem to find the limit. / Gebruik die knypstangstelling om die limiet te bepaal.

$$\lim_{x \rightarrow 0^+} \sqrt{x} e^{\cos 2x}$$

$$-1 \leq \cos 2x \leq 1$$

$$\Rightarrow e^{-1} \leq e^{\cos 2x} \leq e^1$$

$$\Rightarrow e^{-1} \sqrt{x} \leq \sqrt{x} e^{\cos 2x} \leq e \sqrt{x}$$

$$\text{But } \lim_{x \rightarrow 0^+} e^{-1} \sqrt{x} = 0 \text{ and } \lim_{x \rightarrow 0^+} e \sqrt{x} = 0$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \sqrt{x} e^{\cos 2x} = 0$$

Question 20 / Vraag 20 (2)

Find the horizontal asymptotes of $f(x) = \frac{1-e^x}{1-2e^x}$.

Bepaal die horisontale asymptote van $f(x) = \frac{1-e^x}{1-2e^x}$.

$$* \lim_{x \rightarrow \infty} \frac{1-e^x}{1-2e^x} \xrightarrow{\left(\frac{\infty}{\infty}\right)} \lim_{x \rightarrow \infty} \frac{\frac{1}{e^x} - 1}{\frac{1}{e^x} - 2} = \frac{1}{2}$$

$$\Rightarrow y = \frac{1}{2} \text{ is a HA}$$

$$\lim_{x \rightarrow -\infty} \frac{1-e^x}{1-2e^x} = 1$$

$$\Rightarrow y = 1 \text{ is a HA}$$

* With L'Hospital:

$$\lim_{x \rightarrow \infty} \frac{1-e^x}{1-2e^x} \xrightarrow{\left(\frac{\infty}{\infty}\right)} \frac{-e^x}{-2e^x} = \frac{1}{2} \Rightarrow y = \frac{1}{2} \text{ is a HA}$$

$\left(\frac{\infty}{\infty}\right) \downarrow$

Question 21 / Vraag 21 (1, 1, 3)

Consider the functions f , f' and f'' . / Beskou die funksies f , f' en f'' .

$$f(x) = \ln(x^2 + 1)$$

$$f'(x) = \frac{2x}{x^2 + 1}$$

$$f''(x) = \frac{2 - 2x^2}{(x^2 + 1)^2}$$

21.1 Determine the interval(s) on which f is increasing.

Bepaal die interval(le) waarop f styg.

$f'(x) = 0 \Rightarrow x = 0 \Rightarrow x = 0$ is a critical number



A horizontal number line with a vertical tick mark at 0. To the left of 0, there are four minus signs. To the right of 0, there are four plus signs. The label f' is at the right end of the line.

f is increasing on $(0, \infty)$ (or $[0, \infty)$)

21.2 Determine the local extremes of f , if they exist.

Bepaal die lokale ekstreme van f , indien dit bestaan.

f has a local minimum in $x = 0$

The local minimum is $f(0) = \ln 1 = 0$.

21.3 Find the interval of concavity and the inflection points of f , if they exist.

Bepaal die intervalle van konkawiteit en die infleksiepunte van f , indien dit bestaan.

$$\begin{aligned} f''(x) = 0 &\Rightarrow 2 - 2x^2 = 0 \\ &\Rightarrow 2(1 - x^2) = 0 \\ &\Rightarrow x = \pm 1 \end{aligned}$$



A horizontal number line with vertical tick marks at -1 and 1. To the left of -1, there are four minus signs. Between -1 and 1, there are four plus signs. To the right of 1, there are four minus signs. The label f'' is at the right end of the line.

f is concave up on $(-1, 1)$

f is concave down on $(-\infty, -1)$ and on $(1, \infty)$

f has inflection points $(-1, \ln 2)$ and $(1, \ln 2)$

Question 22 / Vraag 22 (3)

Prove that if f is differentiable at $x = a$, then f is continuous in the point $x = a$.

Bewys dat as f differensieerbaar is in $x = a$, dan is f kontinu in die punt $x = a$.

Given: $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists

To prove: $\lim_{x \rightarrow a} f(x) = f(a)$

Proof:

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} (f(x) - f(a)) + f(a)$$

$$= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot (x - a) + f(a)$$

$$= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \rightarrow a} (x - a) + f(a)$$

$f'(a)$ is a finite number

$$= 0 + f(a)$$

$$= f(a)$$

Question 23 / Vraag 23 (3)

Use the Closed Interval Method to find the extreme values of $f(x) = \ln(1+x+x^2)$ on the interval $[-1, 1]$.

Gebruik die Geslote Intervalmethode om die ekstreemwaardes van $f(x) = \ln(1+x+x^2)$ op die interval $[-1, 1]$ te bepaal.

Critical numbers:

$$f'(x) = \frac{1+2x}{1+x+x^2} = 0 \Rightarrow x = -\frac{1}{2}$$

always positive

$$f(-1) = \ln 1 = 0$$

$$f\left(-\frac{1}{2}\right) = \ln \frac{3}{4} < 0$$

$$f(1) = \ln 3$$

$f(1) = \ln 3$ is the absolute maximum of f

$f\left(-\frac{1}{2}\right) = \ln \frac{3}{4}$ is the absolute minimum of f

Bonus question / Bonusvraag (1)

Let f be a function with $f > 0$ and suppose $g = \frac{1}{f}$. Show that if

1. f is increasing, then g is decreasing.
2. f is concave down, then g is concave up.

Laat f 'n funksie wees met $f > 0$ en veronderstel $g = \frac{1}{f}$. Wys dat as

1. f stygend is, dan is g dalend.
2. f konkav na onder is, dan is g konkav na bo.

$$1. \quad g' = \frac{-f'}{f^2}$$

$$\text{But } (f'') > 0 \text{ and } f' > 0 \Rightarrow g' < 0$$

$\Rightarrow g$ is decreasing

$$2. \quad g'' = \frac{-f'' \cdot f^2 + f' \cdot 2f \cdot f'}{f^4} = \frac{-f'' f^2 + 2f (f')^2}{f^4}$$

> 0

$\Rightarrow g$ is concave up.